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Systems and methods for unsupervised paraphrase mining

Abstract

Disclosed embodiments relate to aligning pairs of sentences. Techniques can include receiving a plurality of sentences; generating a graph for each of at least two sentences of the plurality of sentences, wherein generating a graph for each sentence of the at least two sentences comprises: identifying one or more tokens for the sentence; and connecting via edges the one or more tokens; generating a combined graph for the at least two sentences wherein generating a combined graph comprises: aligning the identified tokens of the at least two sentences of the plurality of sentences; identifying matching and non-matching tokens between the at least two sentences based on the alignment; and merging matching tokens into a combined graph node.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7937396	12/2010	Pasca	707/753	G06F 40/247
9104750	12/2014	Dhamdhere	N/A	G06F 16/3322
11262978	12/2021	Cohen	N/A	G06F 16/907
2006/0106592	12/2005	Brockett	704/1	G06F 40/237
2008/0046229	12/2007	Maskey	704/2	G06F 40/289
2009/0119090	12/2008	Niu	704/1	G06F 40/247
2011/0320185	12/2010	Broshi	704/4	G06F 40/44
2016/0070786	12/2015	Myslinski	709/204	G06F 16/3334
2017/0046625	12/2016	Takaai	N/A	G06F 16/3329
2018/0150459	12/2017	Farid	N/A	G06F 16/24578
2018/0181652	12/2017	Sun	N/A	G06F 16/954
2018/0189385	12/2017	Sun	N/A	G06F 40/30
2018/0329883	12/2017	Leidner	N/A	G06F 40/30
2019/0115008	12/2018	Jiang	N/A	G10L 15/22
2019/0268297	12/2018	Vendrow	N/A	H04W 4/12
2019/0295537	12/2018	Sapugay	N/A	G10L 15/19
2019/0370336	12/2018	Prakash	N/A	G06F 40/40
2021/0064657	12/2020	Gopalan	N/A	G06F 18/24
2021/0397793	12/2020	Li	N/A	G06F 40/166
2023/0214413	12/2022	Kugimoto	707/769	G06Q 50/01

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
108829894	12/2017	CN	N/A
2010033462	12/2009	JP	N/A
2012256282	12/2011	JP	N/A
2015045904	12/2014	JP	N/A
2019117484	12/2018	JP	N/A
20180054308	12/2017	KR	N/A

OTHER PUBLICATIONS

Regina Barzilay and Lillian Lee. 2003. Learning to Paraphrase: An Unsupervised Approach Using Multiple-Sequence Alignment. In Proceedings of the 2003 Human Language Technology Conference of the North American Chapter of the Association for Computational L (Year: 2003). cited by examiner

Qader, R., Lecorvé, G., Lolive, D., & Sébillot, P. (2018). Disfluency insertion for spontaneous tts: Formalization and proof of concept . In Statistical Language and Speech Processing: 6th International Conference, SLSP 2018, Mons, Belgium, Oct. 15-16, 2018, Proceedings 6 (pp. 32-44). (Year: 2018). cited by examiner

Vigneshvaran, P., Jayabalan, E., & Kathiravan, A. V. (Mar. 2014). An Eccentric Approach for Paraphrase Detection Using Semantic Matching and Support Vector Machine. In 2014 International Conference on Intelligent Computing Applications (pp. 431-434). IEEE. (Year: 2014). cited by examiner

Japanese Search Report, corresponding with a Japanese Patent Application No. 2020187645, mailed on May 22, 2024. (51 pages). cited by applicant

Notice of Reasons for Refusal, corresponding with a Japanese Patent Application No. 2020187645, mailed on Jun. 28, 2024. (10 pages). cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This is a continuation of U.S. patent application Ser. No. 17/008,563, filed Aug. 31, 2020, which is incorporated herein by reference in its entirety.

BACKGROUND

(1) Improvement of natural language processing systems to allow computers to respond to natural languages is a challenging task. The task becomes increasingly difficult when machines attempt to understand languages and phrases specific to domains and based on limited amounts of data. There is a need for techniques and systems which can easily adapt to domain specific information to respond to the needs of modern natural language systems.

SUMMARY

(2) Certain embodiments of the present disclosure relate to a non-transitory computer readable storage medium storing instructions that are executable by a paraphrase mining system that includes one or more processors to cause the paraphrase mining system to perform a method for aligning pairs of sentences. The method can include receiving a plurality of sentences, generating a graph for each of at least two sentences of the plurality of sentences, wherein generating a graph for each sentence of the at least two sentences comprises: identifying one or more tokens for the sentence, and connecting via edges the one or more tokens, generating a combined graph for the at least two sentences wherein generating a combined graph comprises: aligning the identified tokens of the at least two sentences of the plurality of sentences, identifying matching and non-matching tokens between the at least two sentences based on the alignment, and merging matching tokens into a combined graph node.

(3) According to some disclosed embodiments, generating a combined graph for the at least two sentences can further include determining compatibility among the at least two sentences, and

- removing sentences from the at least two sentences based on the compatibility determination.
- (4) According to some disclosed embodiments, determining compatibility among the at least two sentences can include determining injectivity among the at least two sentences, determining monotonicity among the at least two sentences, and determining transitivity among the at least two sentences.
- (5) According to some disclosed embodiments, determining monotonicity among the at least two sentences can further include determining a consistency among the at least two sentences, and removing sentences from the at least two sentences based on the consistency determination.
- (6) According to some disclosed embodiments, receiving a plurality of sentences can further include determining an intent of each sentence in a set of the plurality sentences, and grouping each sentence in the set of the plurality of sentences into clusters wherein sentences in a cluster share the intent.
- (7) According to some disclosed embodiments, generating a combined graph for the at least two sentences can further include determining an intent of tokens in the at least two sentences, and identifying a set of non-matching tokens between the at least two sentences based on the alignment wherein the non-matching tokens in the set share the intent.
- (8) According to some disclosed embodiments, the non-matching tokens can be paraphrases with one or more words that can be used interchangeably.
- (9) According to some disclosed embodiments, non-aligned tokens can be phatic expressions.
- (10) According to some disclosed embodiments, the phatic expressions in each sentence of the two or more sentences can be removed before generating a combined graph.
- (11) According to some disclosed embodiments, the phatic expressions can be added to the combined graph notation of the two more sentences after aligning the identified tokens.
- (12) According to some disclosed embodiments, the method can further include generating a new sentence from the matching tokens and a subset of the non-matching tokens.
- (13) According to some disclosed embodiments, the new sentence is generated as a response to a question.
- (14) Certain embodiments of the present disclosure relate to computer implemented method for aligning pairs of sentences. The method can include receiving a plurality of sentences, generating a graph for each of at least two sentences of the plurality of sentences, wherein generating a graph for each sentence of the at least two sentences comprises: identifying one or more tokens for the sentence, and connecting via edges the one or more tokens, generating a combined graph for the at least two sentences wherein generating a combined graph comprises: aligning the identified tokens of the at least two sentences of the plurality of sentences, identifying matching and non-matching tokens between the at least two sentences based on the alignment, and merging matching tokens into a combined graph node.
- (15) According to some disclosed embodiments, generating a combined graph for the at least two sentences can further include determining compatibility among the at least two sentences, and removing sentences from the at least two sentences based on the compatibility determination.
- (16) According to some disclosed embodiments, determining compatibility among the at least two sentences can include determining injectivity among the at least two sentences, determining monotonicity among the at least two sentences, and determining transitivity among the at least two sentences.
- (17) According to some disclosed embodiments, generating a combined graph for the at least two sentences can further include determining a consistency among the at least two sentences, and removing sentences from the at least two sentences based on the consistency determination.
- (18) Certain embodiments of the present disclosure relate to a paraphrase mining system. The paraphrase mining system can include one or more memory devices storing processor executable instructions, and one or more processors configured to execute the instructions to cause the paraphrase mining system to perform operations. The operations can include receiving a plurality

of sentences, generating a graph for each of at least two sentences of the plurality of sentences, wherein generating a graph for each sentence of the at least two sentences comprises: identifying one or more tokens for the sentence, and connecting via edges the one or more tokens, generating a combined graph for the at least two sentences wherein generating a combined graph comprises: aligning the identified tokens of the at least two sentences of the plurality of sentences, identifying matching and non-matching tokens between the at least two sentences based on the alignment, and merging matching tokens into a combined graph node.

(19) According to some disclosed embodiments, the operations can further include determining an intent of each sentence in a set of the plurality sentences, and grouping each sentence in the set of the plurality of sentences into clusters wherein sentences in a cluster share the intent.

(20) According to some disclosed embodiments, the operations can further include determining an intent of tokens in the at least two sentences.

(21) According to some disclosed embodiments, the operations can further include generating a new sentence from the matching tokens and a subset of the non-matching tokens.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate several embodiments and, together with the description, serve to explain the disclosed principles. In the drawings:

(2) FIG. 1 is a block diagram showing various exemplary components of a paraphrase mining system for generating word alignment graphs, consistent with embodiments of the present disclosure.

(3) FIG. 2 is a flowchart showing an exemplary method for pre-processing input sentences, consistent with embodiments of the present disclosure.

(4) FIG. 3 is a flowchart showing an exemplary method for building a word alignment graph of a set of sentences, consistent with embodiments of the present disclosure.

(5) FIG. 4 is a block diagram of an exemplary computing device, consistent with embodiments of the present disclosure.

(6) FIG. 5 is a flowchart showing an exemplary method for word alignment, consistent with embodiments of the present disclosure.

(7) FIG. 6A-C are examples of word alignment, consistent with embodiments of the present disclosure.

(8) FIG. 7A illustrates an exemplary paraphrase mining system, consistent with embodiments of the present disclosure.

(9) FIG. 7B illustrates an exemplary paraphrase mining system, consistent with embodiments of the present disclosure, according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

(10) In the following detailed description, numerous details are set forth to provide a thorough understanding of the disclosed example embodiments. It is understood by those skilled in the art that the principles of the example embodiments may be practiced without every specific detail. The embodiments disclosed are exemplary and are intended to disclose every possible embodiment consistent with the claims and disclosure. Well-known methods, procedures, and components have not been described in detail so as not to obscure the principles of the example embodiments. Unless explicitly stated, the example methods and processes described herein are neither constrained to a particular order or sequence nor constrained to a particular system configuration. Additionally, some of the described embodiments or elements thereof can occur or be performed simultaneously, at the same point in time, or concurrently.

(11) As used herein, unless specifically stated otherwise, the term “or” encompasses all possible combinations, except where infeasible. For example, if it is stated that a component may include A or B, then, unless specifically stated otherwise or infeasible, the component may include A, or B, or A and B. As a second example, if it is stated that a component may include A, B, or C, then, unless specifically stated otherwise or infeasible, the component may include A, or B, or C, or A and B, or A and C, or B and C, or A and B and C.

(12) Reference will now be made in detail to the disclosed embodiments, examples of which are illustrated in the accompanying drawings. Unless explicitly stated, sending and receiving as used herein are understood to have broad meanings, including sending or receiving in response to a specific request or without such a specific request. These terms thus cover both active forms, and passive forms, of sending and receiving.

(13) The embodiments described herein provide technologies and techniques for evaluating data sources to identify alignment and mining of paraphrases for improving natural language processing by computing systems. These technologies can use information relevant to the specific domain and application of a data model to prioritize available data or potential data sources. FIGS. 7A-7B describe various data sources accessed or chosen to utilize based on the results of various systems described below. The described technologies can synthesize the data into a coherent data model, that can be used to analyze and compare various paths or courses of action.

(14) These technologies can efficiently evaluate data sources and data, prioritize their importance based on domain and circumstance specific needs, and provide effective and accurate predictions that can be used to evaluate potential courses of action. The technologies and methods allow for the application of data models to personalized circumstances. These methods and technologies allow for detailed evaluation that can improve decision making on a case by case basis.

(15) The described embodiments provide a distinct advantage over existing techniques of natural language processing. Unlike most other processes, the graph-based system described in the disclosed embodiments operate in an unsupervised manner. Additionally, the described embodiments can provide useful insights even with small amounts of data or data that is tied to a particular domain which provides distinct advantages over statistical methods that only work on general purpose phrases and require significantly larger data sets to be effective. By focusing on generating consistent groupings of sentences that share common intent, the embodiments disclosed herein can effectively identify interchangeable phrases sharing the same topic or intent. The ability to do this can provide significant advantages in natural language processing systems that must respond to different individuals or questions that often ask the same thing but in different ways. By allowing for unsupervised, efficient identification of domain specific phrases, the embodiments disclosed herein can provide an improved ability to use natural language processing in various industries and particularized contexts without the need for extensive customization and modification.

(16) FIG. 1 is a block diagram showing various exemplary components of paraphrase mining system **100** for generating word alignment graphs, consistent with embodiments of the present disclosure. The paraphrase mining system **100** can comprise an input sentences store **110** that can be pre-populated using a corpus of sentences. In some embodiments, the input sentences store **110** saves a set of input sentences supplied by a user before passing them to other components of paraphrase mining system **100**. In some other embodiments, the sentences in the input sentences store **110** can be supplied by separate system. In some embodiments, input sentences store can include sentences supplied by user input, other systems, other data sources, or feedback from paraphrase mining system **100** or its components. Input sentences store **110** can be a Relational Database Management System (RDBMS) (e.g., Oracle Database, Microsoft SQL Server, MySQL, PostgreSQL, or IBM DB2). An RDBMS can be designed to efficiently return data for an entire row, or record, from the database in as few operations as possible. An RDBMS can store data by serializing each row of data in a data structure. In an RDBMS, data associated with a record can be

stored serially such that data associated with all categories of the record can be accessed in one operation. Moreover, an RDBMS can efficiently allow access of related records stored in disparate tables. For example, in an RDBMS, tables can be linked by a referential column and the RDBMS can join tables together to retrieve data for a data structure. In some embodiments, the input sentences store **110** can be a non-relational database system (NRDBMS) (e.g., XML, Cassandra, CouchDB, MongoDB, Oracle NoSQL Database, FoundationDB, or Redis). A non-relational database system can store data using a variety of data structures such as, among others, a key-value store, a document store, a graph, and a tuple store. For example, a non-relational database using a document store could combine all of the data associated with a particular identifier into a single document encoded using XML. The input sentences store **110** can also be a memory database such as Memcached. In some embodiments, the contents of input sentences store **110** may exist both in a persistent storage database and in an in-memory database, such as is possible in Redis.

(17) In natural language processing systems, such as paraphrase mining system **100**, paraphrases can be different words or groupings of words that have a similar meaning. Paraphrase mining system **100** can identify paraphrases in the input sentences using an unsupervised, graph-based system, or process. Paraphrase mining system **100** can align words from input sentences in input sentences store **110** (e.g., using pre-processor **120**, word-aligner **130**, or both, as described in more detail below). Using the aligned words and phrases, paraphrase mining system **100** can further generate a word-alignment graph (e.g., using word-alignment graph generator **140**, described in more detail below) to identify phrases and words that can be used interchangeably. The identified paraphrases can be used interchangeably in creating new sentences. The identification of paraphrases can also aid in dropping or eliminating certain phrases from existing sentences without impacting the meaning of those sentences. Such phrases have no paraphrases in other sentences and thus may be regarded as non-essential for the meaning of the sentence and dropped when preparing sentences. By identifying paraphrases in this way, paraphrase mining system **100** can also identify domain-specific paraphrases that would not otherwise be identified through existing statistical methods focused on identifying general-purpose paraphrases. Moreover, paraphrase mining system **100**, outlined above, and described in more detail below, can generate paraphrases from data sets that are generally considered too small for use in existing systems.

(18) As illustrated in FIG. 1, paraphrase mining system **100** can include pre-processor **120**. Pre-processor **120** can process input sentences in input sentences store **110** prior to aligning the words and phrases in the sentences. In some embodiments, pre-processor **120** receives some or all of the input sentences directly instead of loading them from input sentences store **110**. Pre-processor **120** can skip processing certain input sentences before passing them to the word aligner **130** based on predefined criteria. The predefined criteria may include sentences longer than a certain length. The length can be determined by the number of characters, words, or phrases. In some embodiments, the length can be determined by the amount of screen space taken to present the sentence. In some embodiments, the pre-defined criteria can vary with the language of the input sentences. Predefined criteria can also be excluding grammatically incorrect sentences, including incomplete sentences or misspellings, missing punctuation, or sentence abnormalities. The predefined criteria can be set utilizing a graphical user interface. The predefined criteria can be set individually by each user of the system or by the administration of the system. The predefined criteria can be automatically determined based on the input sentences corpus or the language of the input sentences. In some embodiments, pre-processor **120** can store the processed input sentences in input sentences store **110** for later use.

(19) Pre-processor **120** can prepare input sentences for further processing. The processing by pre-processor **120** can include replacing numbers and named entities with symbols or tokens. For example, numbers can be replaced by “NUM” and named entities can be replaced with “ORG.” Pre-processor **120** can further generate the relationship between the processed input sentence and the original sentence. In some embodiments, pre-processor **120** can identify the relationship

between a group of input sentences. Pre-processor **120** can also store the identified relationship between a group of input sentences in the input sentences store **110**. A relationship can be maintained in a hash table with the replacing symbol or token as the key mapping to the named entity, which may act as the value. The hash tables maintaining relationships can be stored in persistent storage, such as the input sentences store **110**. The relationships can be stored in a separate database or tables from the database or tables storing input sentences.

(20) The paraphrase mining system **100** can include word aligner **130** to align words between sentences. Word aligner **130** can receive input sentences from pre-processor **120**. In some embodiments, word aligner **130** can directly access the input sentences store **110** to retrieve a set of input sentences for alignment. Word aligner **130** can select an input sentence from the input sentences store **110** based on defined requirements. Method **300** discussed below describes some ways for selecting input sentences for alignment. Word aligner **130** can operate on a pair of sentences and generate predicted mappings between the tokens of those sentences. Word aligner **130** can result in sets of aligned and non-aligned words that can then be used for further analysis. In some embodiments, word aligner **130** can iterate through all possible sentence pairs to find proper alignment between the set of input sentences.

(21) Word aligner **130** can divide the input sentences into tokens prior to the alignment. Word aligner **130** can align the tokens identified in the input sentences. Word aligner **130** can use word embeddings to identify an alignment between identified tokens. Word embeddings can refer to a learned representation of words that have the same or similar meanings. Word embeddings can include glove embeddings. In some embodiments, the word embeddings may be contextual embeddings that are computed using a neural network architecture such as, for example, BERT and ELMO, etc. In some embodiments, the word aligner **130** can also use general-purpose paraphrases and lexical resources to identify alignment between the identified tokens.

(22) Word aligner **130** can align a pair of input sentences if they have a certain number of tokens within them that can align. Word aligner **130** can also consider the proximity between tokens when aligning sentences. The proximity between the tokens can be based on word embeddings associated with the tokens in the input sentences. The threshold for the number of tokens expected to align or the allowed proximity level between the tokens can be set in the paraphrase mining system **100**. The levels can be updated upon every alignment request by a user or periodically by paraphrase mining system **100**. In some embodiments, the threshold is chosen based on a determined quality of the input sentences or the purpose of aligning input sentences. Accordingly, the target use case for paraphrase mining system **100** can affect the threshold value.

(23) Paraphrase mining system **100** can also include word alignment graph generator **140** for creating directed acyclic graphs (DAGs) of all aligned sentences. The word aligner **130** may supply all aligned sets of input sentences to word alignment graph generator **140**. In some embodiments, combinations of sentences with minimum word alignment can be withheld from word alignment graph generator **140**. The minimum word alignment is configurable. Word alignment graph generator can create a new DAG using an existing DAG and a set of word-aligned input sentences. Word alignment graph generator **140** can retrieve existing DAGs from word alignment DAG store **150**. Word alignment DAG store **150** can store newly created DAGs generated by word alignment graph generator **140** as well as DAGs created by other processes or iterations of paraphrase mining system **100**. Word alignment DAG store **150** may be a database or a flat file system. As described above in reference to input sentences store **110**, word alignment DAG store **150** can be an RDBMS, an NRDBMS, or other type of data store. In some embodiments, the word alignment DAG store **150** may be stored on the same database as input sentences store **110**.

(24) In some embodiments, paraphrase mining system **100** may translate the input sentences to a uniform language prior to the alignment of sentences. Pre-processor **120** can transform the input sentences. Paraphrase mining system **100** can detect the language of the system to determine if translation is required. In some embodiments, the word aligner **130** can align sentences in different

languages using word embeddings available in different languages. Word aligner **130** can request that pre-processor **120** process parts of the input sentence or sentences prior to alignment. Word aligner **130** can request that pre-processor **120** process a pair of sentences. In some embodiments, word aligner **130** requests this alignment even if the sentences do not meet the alignment thresholds for word aligner **130**. The components of the paraphrase mining system **100** may interact with one or more other components of paraphrase mining system **100** or other systems to determine the best alignment of the input sentences. The alignment process for the input sentences is discussed in detail below in reference to method **500**.

(25) The components of paraphrase mining system **100** can run on a single computer or may be distributed across multiple computers or processors. The different components of paraphrase mining system **100** can communicate over a network (e.g., LAN or WAN) or the Internet. In some embodiments, each component can run on multiple computer instances or processors. The instances of each component of the paraphrase mining system **100** can be a part of a connected network such as a cloud network (e.g., Amazon AWS, Microsoft Azure, Google Cloud). In some embodiments, some, or all, of the components of paraphrase mining system **100** are executed in virtualized environments such as a hypervisor or virtual machine.

(26) FIG. **2** is a flowchart showing an exemplary method **200** for pre-processing input sentences, according to embodiments consistent with the present disclosure. The steps of method **200** can be performed by the paraphrase mining system **100** for purposes of illustration. It is appreciated that the illustrated method **200** can be altered to modify the order of steps and to include additional steps.

(27) Method **200** begins at step **201**. In step **210**, pre-processor **120** can group input sentences into clusters. Pre-processor **120** can consider grouping only when the number of input sentences is above a specified threshold number. In some embodiments, the paraphrase mining system **100** sets the threshold for grouping sentences based on the available computation capacity. The threshold can also depend on the total number of words of the group of sentences and the average number of words per input sentence. Pre-processor **120** can group the sentences into compatible groups capable of being aligned. The compatibility of the sentences can depend on the number of similar words between input sentences of the compatible group. The pre-processor **120** can store the clusters of input sentences in input sentences store **110**. The paraphrase mining system **100** can request new input sentences to be grouped with an existing cluster of input sentences stored in input sentences store **110**. In some embodiments, pre-processor **120** can group a new sentence with previously prepared cluster of input sentences only if it is compatible with sentences in the cluster. Otherwise, the new sentence can form a new cluster. An input sentence can be compatible with one or more groups of sentences. A single sentence can be itself a cluster forming a singleton cluster. Such singleton clusters may not need any further processing. Pre-processor **120** may calculate a compatibility score of each input sentence with a group to determine the best compatible group. In some embodiments, the compatible group identification results in splitting the previously formed cluster of input sentences into one or more new groups of sentences.

(28) In step **220**, pre-processor **120** can clean the sentences by removing the phatic expressions. Phatic expressions are those parts of the sentences which do not add additional meaning to the sentence but are only meant as a social function. For example, “Excuse me” and “All right” may be individual sentences or parts of sentences included before or after the actual sentence. The phatic expressions can be identified by identifying similar sentences in input sentences store **110** that exclude the words forming the phatic expression. In some embodiments, the phatic expressions appear before the aligned tokens of two sentences signifying their social function. For example, if the two input sentences are “All right. I will get it done by Tuesday evening,” and “I will complete it by Tuesday evening” pre-processor **220** can identify the phatic expression “All right” and exclude it before the alignment of the two sentences. In some embodiments, the paraphrase mining system **100** maintains a list of phatic expressions. Phatic expressions identified in step **220** may be

added to the list of phatic expressions. Paraphrase mining system **100** can store the list of phatic expressions in input sentences store **110**.

(29) In step **230**, pre-processor **120** of paraphrase mining system **100** can identify nouns in the input sentences and replace them with a standard symbol in order to aid in the alignment of input sentences conveying the same intent in references to different named entities. For example, two input sentences, “Paris is one of the most visited cities in Europe,” and “Tokyo is one of the most visited cities in Asia” could be transformed by replacing “Paris,” “Europe,” “Tokyo,” and “Asia” with a standard symbol such as “<PLACE>.” Replacing nouns with symbols can aid in aligning the two input sentences, which convey similar meaning but refer to different places. In some embodiments, paraphrase mining system **100** can use the same symbol for different kinds of nouns (e.g., places, animals, objects, etc.). Pre-processor **120** can also replace the numbers in an input sentence with a symbol. Pre-processor **120** can store a table of replaced named entities and numbers in input sentences in a table or data store, such as an RDBMS or NRDBMS like those described above. Replacing named entities and numbers in input sentences can avoid considering named entities as paraphrases and, thus, generating incorrect factual sentences from the aligned paraphrases. For example, alignment of sentences “Paris is one of the most visited cities in Europe,” and “Tokyo is one of the most visited cities in Asia” without replacement of named entities can result in the alignment of “Paris” and “Tokyo,” which would allow generating a new sentence, “Tokyo is one of the most visited cities in Europe.” The process for generating new sentences is discussed further below in relation to the description of FIG. 7A. Pre-processor **120**, upon completion of step **230**, can complete (step **299**) the execution of method **200**

(30) FIG. 3 is a flowchart showing an exemplary method **300** for building a word alignment graph, consistent with embodiments of the present disclosure. The steps of method **300** can be performed by paraphrase mining system **100** for the purposes of illustration. It is appreciated that the illustrated method **300** can be altered to modify the order of steps and to include additional steps.

(31) Method **300** begins at step **301**. In step **310**, word alignment graph generator **140** can group input sentences into compatible groups in order to aid with the alignment and extraction of paraphrases. Word alignment graph generator **140** can determine compatibility between a group of sentences based on various rules. These rules can be based on principles of injectivity, monotonicity, and transitivity.

(32) Word alignment graph generator **140** can determine that two sentences have satisfied injectivity if each word in the first sentence maps to only one word in the second sentence and vice versa. Injectivity does not restrict a first sentence from having additional words which are not mapped to the words in the second sentence and vice versa.

(33) Word alignment graph generator **140** considers satisfaction of monotonicity if the order of mapping between words in a sentence matches the order of the words in other sentences. For example, if word **w1** in a first input sentence appears before word **w2** in a second input sentence, then the word **w1** needs to map to a word before **w2** in the second input sentence. Because of the structure or phrasing of some input sentences, the sentences can result in mapped words that appear in different orders among the sentences even though they have the same intent or meaning. Word alignment graph generator **140** can identify these inconsistencies and remove the sentences from the group to ensure they do not disrupt the graph creation. Sentences removed from a group for failing to satisfy monotonicity may be added to a new group or another existing group. By removing these sentences, word alignment graph generator **140** can ensure the consistency of the sentences.

(34) Word alignment graph generator **140** considers satisfaction of transitivity requirement when multiple sentences map to the same words among themselves. For example, if **w1** in a first input sentence maps to **w2** in a second input sentence and **w2** in the second input sentence maps to **w3** in a third input sentence, then **w1** must either have no mappings in the third sentence or it must also map to **w3** to be considered transitive. If not, the sentences are not considered transitive and,

therefore, can be considered incompatible. Similar to injectivity, monotonicity and transitivity requirements do not require that every word in each sentence is mapped to words in other compatible sentences.

(35) If the above requirements, injectivity, monotonicity, and transitivity are present in a set of sentences, word alignment graph generator **140** can consider the sentences compatible. These requirements can ensure that the resulting graph generated by word alignment graph generator **140** is compact or forms a DAG.

(36) Word alignment graph generator **140** allows a single input sentence to be part of only one compatible group. When a set of sentences is considered compatible, they can be removed from the set of available sentences. The compatibility of a sentence is determined across all available sentences. In some embodiments, the compatible groups are updated based on new sets of input sentences.

(37) In step **320**, word alignment graph generator **140** can create DAGs of compatible sentences by first transforming each sentence into a line graph. The transformation of an input sentence into a line graph can require splitting the sentence into tokens identified by word aligner **130** and then creating nodes for each of the identified tokens. The tokens can include certain words replaced by symbols, as discussed above in reference to method **200**. An example line graph is shown in FIG. **6A** and described in more detail below.

(38) A DAG of two input sentences in a compatible group can be constructed by first identifying the common tokens between the two input sentences. Tokens in a sentence that have no matching tokens in the other sentence are considered paraphrases that can be used interchangeably. In some embodiments, a sequence of tokens (also called a SPAN) in each sentence that does not have a match can be considered paraphrases that can be used interchangeably. Word alignment graph generator **140** can also reapply the rules for compatibility while aligning sentences.

(39) In step **330**, word alignment graph generator **140** combines various DAGs within a compatible group generated in step **320** into a single, combined DAG. Word alignment graph generator **140** can include a start and stop node before at the beginning and end, respectively, of the graph. Word alignment graph generator **140**, upon completion of step **300**, completes (step **399**) the execution of method **300**.

(40) FIG. **4** is a block diagram of an exemplary computing device **400**, consistent with embodiments of the present disclosure. In some embodiments, computing device **400** can be a specialized server providing the functionality described herein. In some embodiments, components of paraphrase mining system **100**, such as input sentences store **110**, pre-processor **120**, word aligner **130**, word alignment graph generator **140**, and word alignment DAG store **150** can be implemented using the computing device **400** or multiple computing devices **400** operating in parallel. Further, the computing device **400** can be a second device providing the functionality described herein or receiving information from a server to provide at least some of the described functionality. Moreover, the computing device **400** can be an additional device or devices that store or provide data consistent with embodiments of the present disclosure and, in some embodiments, computing device **400** can be a virtualized computing device such as a virtual machine, multiple virtual machines, or a hypervisor.

(41) Computing device **400** can include one or more central processing units (CPUs) **420** and a system memory **421**. Computing device **400** can also include one or more graphics processing units (GPUs) **425** and graphic memory **426**. In some embodiments, computing device **400** can be a headless computing device that does not include GPU(s) **425** or graphic memory **426**.

(42) CPUs **420** can be single or multiple microprocessors, field-programmable gate arrays, or digital signal processors capable of executing sets of instructions stored in a memory (e.g., system memory **421**), a cache (e.g., cache **441**), or a register (e.g., one of registers **440**). CPUs **420** can contain one or more registers (e.g., registers **440**) for storing various types of data including, inter alia, data, instructions, floating-point values, conditional values, memory addresses for locations in

memory (e.g., system memory **421** or graphic memory **426**), pointers and counters. CPU registers **440** can include special-purpose registers used to store data associated with executing instructions such as an instruction pointer, an instruction counter, or a memory stack pointer. System memory **421** can include a tangible or a non-transitory computer-readable medium, such as a flexible disk, a hard disk, a compact disk read-only memory (CD-ROM), magneto-optical (MO) drive, digital versatile disk random-access memory (DVD-RAM), a solid-state disk (SSD), a flash drive or flash memory, processor cache, memory register, or a semiconductor memory. System memory **421** can be one or more memory chips capable of storing data and allowing direct access by CPUs **420**. System memory **421** can be any type of random-access memory (RAM), or other available memory chip capable of operating as described herein.

(43) CPUs **420** can communicate with system memory **421** via a system interface **450**, sometimes referred to as a bus. In embodiments that include GPUs **425**, GPUs **425** can be any type of specialized circuitry that can manipulate and alter memory (e.g., graphic memory **426**) to provide or accelerate the creation of images. GPUs **425** can have a highly parallel structure optimized for processing large, parallel blocks of graphical data more efficiently than general-purpose CPUs **420**. Furthermore, the functionality of GPUs **425** can be included in a chipset of a special purpose processing unit or a co-processor.

(44) CPUs **420** can execute programming instructions stored in system memory **421** or other memory, operate on data stored in memory (e.g., system memory **421**), and communicate with GPUs **425** through the system interface **450**, which bridges communication between the various components of the computing device **400**. In some embodiments, CPUs **420**, GPUs **425**, system interface **450**, or any combination thereof, are integrated into a single chipset or processing unit. GPUs **425** can execute sets of instructions stored in memory (e.g., system memory **421**), to manipulate graphical data stored in system memory **421** or graphic memory **426**. For example, CPUs **420** can provide instructions to GPUs **425**, and GPUs **425** can process the instructions to render graphics data stored in the graphic memory **426**. Graphic memory **426** can be any memory space accessible by GPUs **425**, including local memory, system memory, on-chip memories, and hard disk. GPUs **425** can enable displaying of graphical data stored in graphic memory **426** on display device **424** or can process graphical information and provide that information to connected devices through network interface **418** or I/O devices **430**.

(45) Computing device **400** can include a display device **424** and input/output (I/O) devices **430** (e.g., a keyboard, a mouse, or a pointing device) connected to I/O controller **423**. I/O controller **423** can communicate with the other components of computing device **400** via system interface **450**. It should now be appreciated that CPUs **420** can also communicate with system memory **421** and other devices in manners other than through system interface **450**, such as through serial communication or direct point-to-point communication. Similarly, GPUs **425** can communicate with graphic memory **426** and other devices in ways other than system interface **450**. In addition to receiving input, CPUs **420** can provide output via I/O devices **430** (e.g., through a printer, speakers, bone conduction, or other output devices).

(46) Furthermore, the computing device **400** can include a network interface **418** to interface to a LAN, WAN, MAN, or the Internet through a variety of connections including, but not limited to, standard telephone lines, LAN or WAN links (e.g., 802.21, T1, T3, 56 kb, X.25), broadband connections (e.g., ISDN, Frame Relay, ATM), wireless connections (e.g., those conforming to, among others, the 802.11a, 802.11b, 802.11b/g/n, 802.11ac, Bluetooth, Bluetooth LTE, 3GPP, or WiMax standards), or some combination of any or all of the above. Network interface **418** can comprise a built-in network adapter, network interface card, PCMCIA network card, card bus network adapter, wireless network adapter, USB network adapter, modem or any other device suitable for interfacing the computing device **400** to any type of network capable of communication and performing the operations described herein.

(47) FIG. 5 is a flowchart showing an exemplary method **500** for paraphrase mining, consistent

with embodiments of the present disclosure. The steps of method **500** can be performed by paraphrase mining system **100** of FIG. **1** executing on or otherwise using the features of computing device **400** of FIG. **4** for purposes of illustration. It is appreciated that the illustrated method **500** can be altered to modify the order of steps and to include additional steps.

(48) In step **510**, paraphrase mining system **100** can receive a plurality of sentences. Paraphrase mining system **100** can receive the input sentences over a network at, for example, network interface **418**. If not received from input sentences store **110**, paraphrase mining system **100** can store the plurality of input sentences in input sentences store **110**, which can be present in, for example, storage **428**. Paraphrase mining system **100** can further pre-process the received input sentences using pre-processor **120** prior to storing them in the input sentences store **110**. The pre-processing steps can include one or more of the steps defined in the method **200** above.

(49) In step **520**, the paraphrase mining system **100** can generate a graph of each of the received input sentences. The graph of an input sentence can be a line graph, as described in method **300** above. The paraphrase mining system **100** can update the named entities and numbers within the input sentence to common symbols for alignment purposes, as described in step **230** of method **200** above.

(50) In step **530**, the paraphrase mining system **100** can combine the graphs of the input sentences to form one or more DAGs. The paraphrase mining system **100** can identify compatible sentences prior to combining them to form DAGs. The compatibility of input sentences can be based on rules described in step **310** of method **300** above (e.g., injectivity, monotonicity, and transitivity). The paraphrase mining system **100** may then combine multiple DAGs to create a combined graph of the compatible group of sentences.

(51) In step **540**, paraphrase mining system **100** can add start and stop nodes to the combined graph of the compatible group of sentences. The start node may be added prior to the first common token between the combined sentences. The stop node may be added after the last common token

(52) In step **550**, paraphrase mining system **100** can mark a compatible group of sentences forming a combined DAG as an aligned set of sentences. Marking a sentence can also include a compatibility score of the sentence in relation to other sentences in the group. In some embodiments, the marking step also includes generating an alignment score. Paraphrase mining system **100** can consider a set of marked sentences for regrouping when a new set of input sentences are introduced. The marking of the aligned sentences can be temporary while the current set of input sentences are being processed.

(53) In step **560**, paraphrase mining system **100** can store the combined graph of two or more sentences generated in step **540** in word alignment DAG store **150**. Paraphrase mining system **100**, upon completion of step **560**, completes (step **599**) executing method **500** on computing device **400**.

(54) FIG. **6A-C** are examples of word alignment of a single sentence, aligned sentences, and a combined group of sentences. The figures show example sentences with tokens represented by nodes in a graph. The edges of a node connect a token to the previous and subsequent tokens in the sentence. FIG. **6A** illustrates a single input sentence “The world has shrugged off the crisis” using a line graph **610**. Each sentence in a compatible group, as identified by word alignment graph generator **140** and described in step **310** of method **300**, can be transformed into a line graph.

(55) FIG. **6B** illustrates a DAG **620** of two input sentences with mapped tokens. The common tokens between the two sentences can be represented using a single set of nodes with one node for each common token. This can be accomplished by combining or merging the nodes representing common or matching tokens into a single, combined graph node while preserving edge connections to other nodes in the graph. The non-common or non-matching tokens between the two input sentences are shown separately as separate nodes. While the individual words within the tokens may not have a common meaning, the paraphrases represented by the non-common tokens can convey the same meaning and thus can be used interchangeably. For example, the paraphrases

“shrugged off the” (e.g., illustrated by nodes **614**, **615** and **616**) and “fully recovered from” (e.g., illustrated by nodes **624**, **625**, and **626**) can be used interchangeably. In some embodiments, a single token may only include a single word. In some embodiments, the sequence of tokens may include common tokens and/or words. The common tokens between the input sentences can match the order and position within the paraphrases. For example, the paraphrases “fully recovered from” (e.g., illustrated by nodes **624**, **625**, and **626**) and “fully recouped from” (not shown in the figure) have matching tokens “fully” and “from” between both paraphrases.

(56) FIG. 6C illustrates a combined DAG **630** of multiple sentences with the identified paraphrases between common nodes **613** and **617**. The combined DAG **630** can include additional tokens (e.g., nodes **637** and **638**) after the last common node **617**. These tokens can be considered as optional tokens to be used in other sentences generated using the identified paraphrases. The additional nodes **637** and **638** and the last common node **617** are connected to the stop node **639**, indicating that the additional tokens can be optional. Similar to FIG. 6B, the phrases between common tokens (e.g. common tokens **613** and **617**) can be used interchangeably. The common tokens include start node **631** and stop node **639**. For example, in DAG **630**, the phrases “crisis” (e.g., represented by node **617**), “crisis already” (e.g., represented by nodes **617** and **637**), and “crisis completely” (e.g., represented by nodes **617** and **638**), can be used interchangeably. In some embodiments, only the non-matching portions of phrases may be evaluated to be either used interchangeably or marked optional. For example, “already” (e.g., represented by node **637**) and “completely” (e.g., represented by node **638**) may be considered optional, and the sentences may be formed without these words. The set of phrases between nodes **613** and **617** and **617** and **639** can be combined to form a new set of sentences not provided to paraphrase mining system **100**. Generation of new sentences using paraphrase mining system **100** is discussed further in relation to FIG. 7A below.

(57) FIG. 7A illustrates an exemplary use case for using a natural language processing system **700** for training a machine learning model, consistent with embodiments of the present disclosure. The use case scenario includes natural language processing system **700** for training a system about one or more languages using user provided and automatically generated sentences.

(58) As illustrated in FIG. 7A, natural language processing system **700** can include word alignment DAG store **150** of the paraphrase mining system **100**. In some embodiments, the paraphrase mining system **100** may be included in the natural language processing system **700**. The paraphrase mining system **100** may be adapted to act as a natural language processing system **700**. Natural language processing system **700** can also include paraphrase miner **710**. Paraphrase miner **710** can identify paraphrases to use to generate new sentences to train the machine learning model **720** of natural language processing system **700**. Machine learning model **720** can communicate with paraphrase generator **711** over network **730**. In some embodiments, machine learning model **720** can be generated on the same computing device (e.g., computing device **400** of FIG. 4) executing paraphrase miner **100**. Natural language processing system **700** can train machine learning model **720** with multiple phrases having similar meaning using paraphrase miner **710**.

(59) The output generated by the word alignment graph generator **140** of the paraphrase mining system **100** can be used as an input to the natural language processing system **700**. Paraphrase mining system **100** can include paraphrase generator **711** and paraphrase filter **712**. In some embodiments paraphrase miner **710** can be separate from paraphrase mining system **100**. Paraphrase generator **711** takes the word aligned DAGs stored in a word alignment DAG store **150** to generate paraphrases. The paraphrase generator may generate a new paraphrase by following a path from a start node to a stop node for the word aligned DAGs. For example, paraphrase generator **711** can use DAG **630** of FIG. 6C as input. Paraphrase generator **711** can begin traversing DAG **630** at start node **631** and follow any of the paths to stop node **639**. The traversal of DAG **630** requires all nodes to be connected by edges, and paraphrase generator **711** can generate all possible phrases by visiting every node of the directed acyclic graph stored in the word alignment DAG store **150**. In some embodiments, the paraphrases generated by paraphrase generator **711** can be

stored in word alignment DAG store **150** or other similar database stores. The generated paraphrases may be stored in the word alignment DAG store **150** in databases or tables separated from the databases and tables used to store the direct acyclic graphs generated by word alignment graph generator **140** and described in method **300** of FIG. **3**. In some embodiments, the paraphrase generator **711** submits the paraphrase or set of paraphrases to a paraphrase filter **712** to review the generated paraphrases. Paraphrase filter **712** can filter phrases that are factually inaccurate as described in step **230** of method **200** of FIG. **2**.

(60) Paraphrase miner **710** can be connected to the paraphrase mining system **100** over network **730**. The paraphrases generated by paraphrase miner **710** can be supplied to the machine learning model **720** over network **730**. In some embodiments paraphrase miner **710** and paraphrase mining system **100** can coexist on the same computing device (e.g., computing device **400** of FIG. **4**) without requiring a network for communication. Paraphrase miner **710** may be a variation of the implemented paraphrase mining system **100**. The paraphrase miner **710** may group and/or store the sentences identified for training the machine learning model **720** as training data source **740**. The training data source **740** may be present in temporary storage or persisted for future use. Several ways of populating training data source **740** are described below.

(61) The described paraphrase generation can be triggered by a request for alignment of new input sentences sent to the paraphrase mining system **100**. In some embodiments, the paraphrase generation can be requested at periodic intervals by natural language processing system **700**.

(62) The natural language processing system **700** can be utilized to generate a corpus of sentences to further train the paraphrase mining system **100**. Given a set of input sentences, the natural language processing system **700** can generate a set of new paraphrased sentences and increase the amount of our training data for the paraphrase mining system **100**. Such a new dataset of sentences can be simply created by mixing and matching phrases from the input sentences already present in the input sentence store **110** and processed by paraphrase mining system **100**. The generated corpus of sentences may be included in the training data source **740** and used to train the machine learning model **720**.

(63) In some embodiments, the natural language processing system **700** may help create a new training dataset (as opposed to enriching the existing dataset with new combinations of paraphrases as described above). In this case, the paraphrases identified using paraphrase mining system **100** may be used to collect more data for the training data source **730** and supplied later to train the machine learning model **720**. For example, if paraphrase mining system **100** discovers that “Uber” can be a paraphrase for “Taxi,” the natural language processing system **700** may go and fetch sentences mentioning “Uber” from the web to train the machine learning model **720** that needs to handle transportation requests. The paraphrases identified by the paraphrase mining system **100** can thus aid in training the machine learning model **720** of natural language processing system **700** in a targeted manner or focus on a certain domain of business.

(64) In some embodiments, the paraphrase mining system **100** (in particular the paraphrase filter **712**) may identify optional phrases (which do not qualify as paraphrases) and clean the input sentences and remove these optional phrases. Such a data cleaning step can help provide higher quality data to machine learning model **720**.

(65) FIG. **7B** illustrates a use case scenario of an exemplary paraphrase mining system **100** interaction by users of user devices. The use case scenario includes a question and answer (“Q & A”) system **750** for responding to questions from user devices using phrases mined by paraphrase mining system **100**.

(66) As illustrated in FIG. **7B**, the Q & A system **750** may include a paraphrase miner **710** to generate responses to the questions. The Q & A system **750** may also include user devices **760** connected to the paraphrase miner **710**, as described in FIG. **7A** above to retrieve phrases that can be used to respond to the questions sent from user devices **760**. The user devices **760** can be connected to the paraphrase miner **710** over a network **730**. Network **730** can be a wired or a

wireless network and can also connect with a paraphrase mining system such as paraphrase mining system **100** of FIG. **1**. In some embodiments, Q & A system **750** can include paraphrase mining system **100**. User devices **760** can connect to paraphrase miner **710** over the Internet. User devices **760** can include, for example, mobile devices **761** and desktop device **762**. Other types of user devices can also be used.

(67) Paraphrase miner **710** can generate paraphrases for responding to questions by accessing word-aligned input sentences stored as directed acyclic graphs (DAGs) in the word alignment DAG store **150** of paraphrase mining system **100**. Paraphrase miner **710** can, among other things, be part of paraphrase mining system **100**, connected to the paraphrase mining system **100** over the network **730**, or communicate directly with paraphrase mining system **100**. In some embodiments, paraphrase mining system **100** can call paraphrase miner **710** to identify the possible paraphrases of the aligned input sentences as part of the alignment process.

(68) Q & A system **750** can use paraphrases generated using paraphrase miner **710** to respond to questions from user devices **760**. The identification of the right response can include alignment of questions from the user devices **760** with sentences generated using new phrases mined by paraphrase miner **710** or paraphrase mining system **100**. The input sentences best aligning with the questions can be used to identify appropriate responses to the questions. The alignment process can follow the same steps described above for method **500** of FIG. **5**. In some embodiments, Q & A system **750** can use features of natural language processing system **700** of FIG. **7A** to further identify potential responses to the questions from user devices **760**. The relationships between the replacing symbols or tokens and the replacing named entities created by pre-processor **120** can be used to retrieve the named entities to be used in responses to questions sent to the Q & A system **750**.

(69) In the Q & A system **750**, a new set of questions with tokens matching the previously responded questions can be identified by the paraphrase mining system **100**. The responses prepared for such questions with matching paraphrases may be utilized to prepare responses to the new questions. The prepared responses can directly use the responses to the previously responded questions or may form part of the new responses prepared for the latest questions.

(70) The exemplary paraphrases mining system **100** can be part of other systems similar to question and answer system **750**. The same Q & A system **750** can be used for different applications. The other applications can include a search engine. For example, the paraphrase mining system **100** that may have identified that the tokens “rentals” and “apartments” are paraphrases that can be used interchangeably, then a search for “cheap rentals in SF” maybe expand by the paraphrase mining system **100** acting as a search engine or part of a search engine to return any document/result that mentions “apartments” as well as “rentals.” Such query expansion and output using the paraphrase mining system **100** can be achieved by finding matching paraphrases.

(71) Example embodiments are described above with reference to flowchart illustrations or block diagrams of methods, apparatus (systems) and computer program products. It will be understood that each block of the flowchart illustrations or block diagrams, and combinations of blocks in the flowchart illustrations or block diagrams, can be implemented by computer program product or instructions on a computer program product. These computer program instructions may be provided to a processor of a computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart or block diagram block or blocks.

(72) These computer program instructions may also be stored in a computer readable medium that can direct one or more hardware processors of a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions stored in the computer readable medium form an article of manufacture including instructions that implement the function/act specified in the flowchart or block diagram block or blocks.

(73) The computer program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other devices to cause a series of operational steps to be performed on the computer, other programmable apparatus or other devices to produce a computer implemented process such that the instructions that execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flowchart or block diagram block or blocks

(74) Any combination of one or more computer readable medium(s) may be utilized. The computer readable medium may be a non-transitory computer readable storage medium. In the context of this document, a computer readable storage medium may be any tangible medium that can contain or store a program for use by or in connection with an instruction execution system, apparatus, or device.

(75) Program code embodied on a computer readable medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber cable, RF, IR, etc., or any suitable combination of the foregoing.

(76) Computer program code for carrying out operations for example embodiments may be written in any combination of one or more programming languages, including an object-oriented programming language such as Java, Smalltalk, C++ or the like and conventional procedural programming languages, such as the "C" programming language or similar programming languages. The program code may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

(77) The flowchart and block diagrams in the figures illustrate examples of the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various embodiments. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams or flowchart illustration, and combinations of blocks in the block diagrams or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

(78) It is understood that the described embodiments are not mutually exclusive, and elements, components, materials, or steps described in connection with one example embodiment may be combined with, or eliminated from, other embodiments in suitable ways to accomplish desired design objectives.

(79) In the foregoing specification, embodiments have been described with reference to numerous specific details that can vary from implementation to implementation. Certain adaptations and modifications of the described embodiments can be made. Other embodiments can be apparent to those skilled in the art from consideration of the specification and practice of the invention disclosed herein. It is intended that the specification and examples be considered as exemplary only. It is also intended that the sequence of steps shown in figures are only for illustrative purposes and are not intended to be limited to any particular sequence of steps. As such, those skilled in the art can appreciate that these steps can be performed in a different order while implementing the same method.

Claims

1. A natural language processing system comprising: one or more memory devices storing processor executable instructions; and one or more processors configured to execute the instructions to cause the system to perform operations comprising: receiving a set of input sentences; generating a set of new paraphrased sentences based on the received set of input sentences, wherein generating the set of new paraphrased sentences comprises: generating a graph for each of at least two input sentences of the set of input sentences by identifying one or more tokens for the at least two input sentences and connecting via edges the one or more tokens; generating a combined graph for the at least two input sentences by aligning the identified one or more tokens to determine matching and non-matching tokens, the non-matching tokens representing one or more interchangeable words, wherein the aligning is based on word embeddings generated by a neural network; and generating the set of new paraphrased sentences based on the determined matching and non-matching tokens; determining a group of compatible sentences from the generated set of new paraphrased sentences based on a first threshold value, wherein a value indicating compatibility between sentences in the group of compatible sentences is above the first threshold value; providing the group of compatible sentences as training data for a machine learning model, wherein the machine learning model is associated with a natural language processing system configured to respond to a user input having a question by aligning the question with the group of compatible sentences; expanding the question using the group of compatible sentences; and providing, based on the expanding, at least one additional output based on the question.
2. The system of claim 1, wherein generating the set of new paraphrased sentences further comprises filtering out factually inaccurate paraphrased sentences from the set of new paraphrased sentences.
3. The system of claim 1, wherein training the machine learning model further comprises: generating additional training data based on a determination of at least one word that is interchangeable for the one or more tokens and accessing online data comprising sentences including the at least one interchangeable word; and training the machine learning model using the online data.
4. The system of claim 1, wherein generating the set of new paraphrased sentences based on the determined matching and non-matching tokens includes following a path from a start node to a stop node in the combined graph.
5. The system of claim 1, wherein generating the set of new paraphrased sentences further comprises: determining compatibility among at least two sentences from the set of new paraphrased sentences; and removing sentences from the set of new paraphrased sentences based on the compatibility determination.
6. The system of claim 1, wherein generating the set of new paraphrased sentences further comprises: determining compatibility among at least one input sentence and at least one sentence from the set of new paraphrased sentences; and removing sentences from the set of new paraphrased sentences based on the compatibility determination.
7. The system of claim 1, wherein receiving the set of input sentences further comprises: determining an intent of each sentence in the set of input sentences; and grouping each sentence in the set of input sentences into clusters wherein sentences in a cluster share the intent.
8. The system of claim 1, wherein generating a combined graph for the at least two of the set of input sentences further comprises: determining an intent of tokens in the at least two input sentences; and identifying a set of non-matching tokens between the at least two input sentences based on the alignment, wherein the non-matching tokens in the set share the intent.
9. The system of claim 8, wherein the set of non-matching tokens includes paraphrases with one or

more words that can be used interchangeably.

10. The system of claim 9, wherein the set of non-matching tokens include phatic expressions.

11. The system of claim 10, wherein the phatic expressions in each sentence of the at least two input sentences are removed before generating a combined graph.

12. The system of claim 11, wherein the phatic expressions are added to the combined graph for the at least two of the set of input sentences after aligning the identified one or more tokens.

13. A method, comprising: receiving a set of input sentences; generating a set of new paraphrased sentences based on the received set of input sentences, wherein generating the set of new paraphrased sentences comprises: generating a graph for each of at least two input sentences of the set of input sentences by identifying one or more tokens for the at least two input sentences and connecting via edges the one or more tokens; generating a combined graph for the at least two input sentences by aligning the identified one or more tokens to determine matching and non-matching tokens, the non-matching tokens representing one or more interchangeable words, wherein the aligning is based on word embeddings generated by a neural network; and generating the set of new paraphrased sentences based on the determined matching and non-matching tokens; determining a group of compatible sentences from the generated set of new paraphrased sentences based on a first threshold value, wherein a value indicating compatibility between sentences in the group of compatible sentences is above the first threshold value; providing the group of compatible sentences as training data for a machine learning model, wherein the machine learning model is associated with a natural language processing system configured to respond to a user input having a question by aligning the question with the group of compatible sentences; expanding the question using the group of compatible sentences; and providing, based on the expanding, at least one additional output based on the question.

14. The method of claim 13, wherein generating the set of new paraphrased sentences further comprises filtering out factually inaccurate paraphrased sentences from the set of new paraphrased sentences.

15. The method of claim 13, wherein training the machine learning model further comprises: generating additional training data based on a determination of at least one word that is interchangeable for the one or more tokens and accessing online data comprising sentences including the at least one interchangeable word; and training the machine learning model using the online data.

16. The method of claim 13, wherein generating a combined graph for the at least two of the set of input sentences further comprises: determining an intent of tokens in the at least two input sentences; and identifying a set of non-matching tokens between the at least two input sentences based on the alignment, wherein the non-matching tokens in the set share the intent.

17. A non-transitory computer readable storage medium storing instructions that are executable by one or more processors to cause a system to perform operations comprising: receiving a set of input sentences; generating a set of new paraphrased sentences based on the received set of input sentences, wherein generating the set of new paraphrased sentences comprises: generating a graph for each of at least two input sentences of the set of input sentences by identifying one or more tokens for the at least two input sentences and connecting via edges the one or more tokens; generating a combined graph for the at least two input sentences by aligning the identified one or more tokens to determine matching and non-matching tokens, the non-matching tokens representing one or more interchangeable words, wherein the aligning is based on word embeddings generated by a neural network; and generating the set of new paraphrased sentences based on the determined matching and non-matching tokens; determining a group of compatible sentences from the generated set of new paraphrased sentences based on a first threshold value, wherein a value indicating compatibility between sentences in the group of compatible sentences is above the first threshold value; providing the group of compatible sentences as training data for a machine learning model, wherein the machine learning model is associated with a natural language

processing system configured to respond to a user input having a question by aligning the question with the group of compatible sentences; expanding the question using the group of compatible sentences; and providing, based on the expanding, at least one additional output based on the question.

18. The medium of claim 17, wherein generating the set of new paraphrased sentences further comprises filtering out factually inaccurate paraphrased sentences from the set of new paraphrased sentences.

19. The medium of claim 17, wherein training the machine learning model further comprises: generating additional training data based on a determination of at least one word that is interchangeable for the one or more tokens and accessing online data comprising sentences including the at least one interchangeable word; and training the machine learning model using the online data.

20. The medium of claim 17, wherein generating a combined graph for the at least two of the set of input sentences further comprises: determining an intent of tokens in the at least two input sentences; and identifying a set of non-matching tokens between the at least two input sentences based on the alignment, wherein the non-matching tokens in the set share the intent.
